

Hewlett-Packard Company DeskJet Printer Case

Instructor: Sairam Sriraman

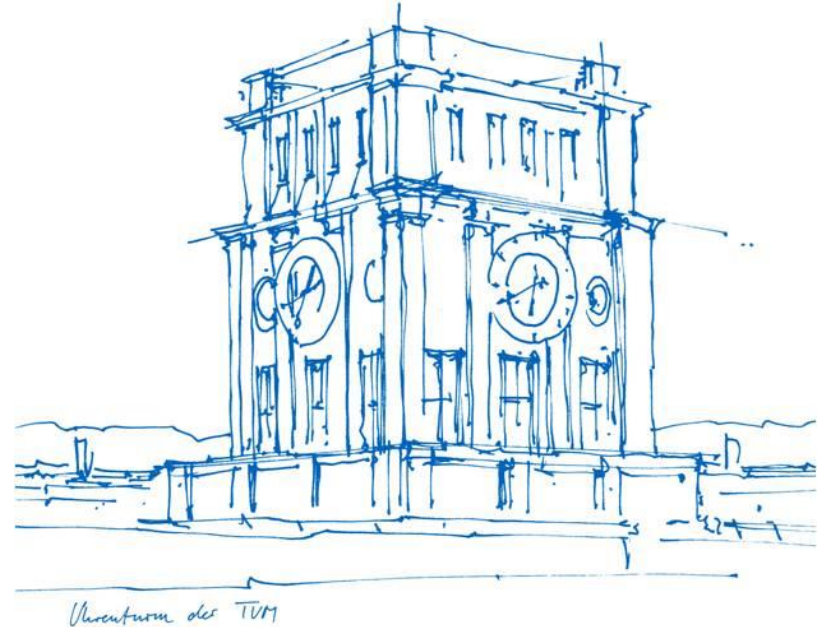
Examiner: Prof. Dr. David Wuttke

Technische Universität München

School of Management

Campus Heilbronn

Heilbronn, 2026



Hewlett-Packard Company DeskJet Printer Supply Chain (A)

Hewlett-Packard Supply Chain Case Questions

1. What is the situation faced by Hewlett-Packard in the early 1990s? Which tradeoffs does the outstanding decision entail?
2. Position HP's deskJet Printer in a product life-cycle model. What are the main characteristics/drivers at this stage?
3. Describe the key performance indicators/metrics that are relevant for HP.
4. Analyze the European demand (a) on a SKU level (i.e., option level) and (b) on an aggregate level.
5. What service level do you consider optimal? What Line Item Fill Rate (LIFR) does this service level imply?
6. Calculate the optimal safety stock for the status quo (hint: use the order-up-to model).
7. Calculate the optimal safety stock for the case where products are shipped via air.
8. Calculate the optimal safety stock for the case where integration is done in Europe (pooled case).
9. Discuss possible alternatives, process implications, and make a recommendation.

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1. Situation of HP DeskJet Printers

Situation and problems faced by HP DeskJet Vancouver division and European DCs

- Sales increase but inventory increases at about the same pace. However, recall the economic order quantity model according to which $Q^* = \sqrt{\frac{2K\lambda}{h}}$, hence, as demand rate λ doubles, inventory should roughly grow by a factor of $\sqrt{2} \approx 1.41$ but should not grow at the same pace. This indicates some inefficiencies.
- Stock-outs are observed while warehouses are filled.
- Competition is becoming stronger reducing margins and requiring better service.
- Production in Vancouver is perfectly lean – but what about distribution centers (DCs) still make-to-stock

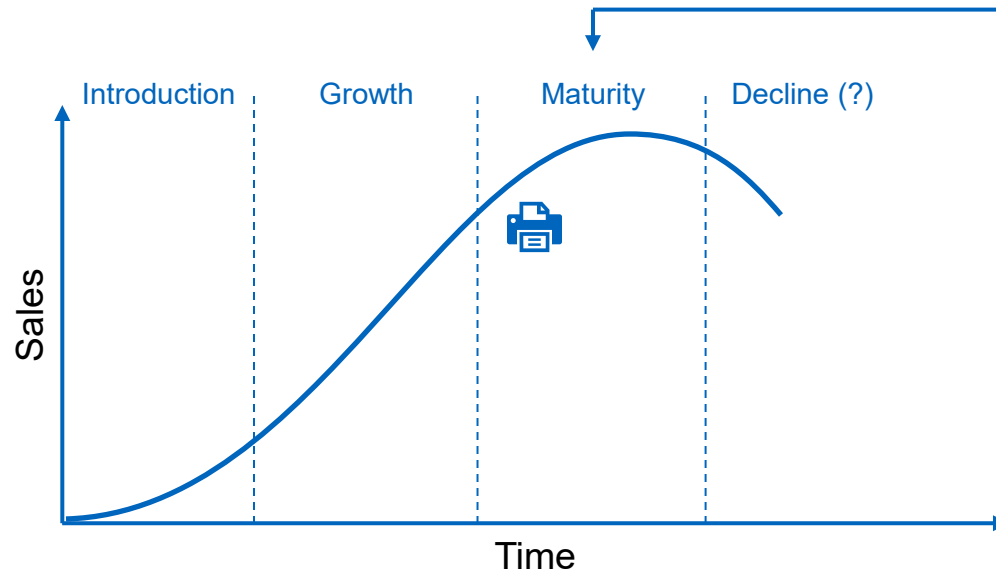
Tradeoffs in ensuing decision

- Service level versus inventory costs
- Longer lead time (ship) versus higher transportation costs (air)
- Highly efficient processes in European DCs with high inventory versus decentralized lean production with skilled (expensive) labor
- Optimality of solution (what do the models tell us?) versus real-life incentives (who would benefit?)

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2. Position of HP DeskJet Printers in Life-Cycle Model



Customer perspective

- Commoditization (less differentiation)
- Lower margins (due to competition)
- Higher customer expectations
- Peak sales

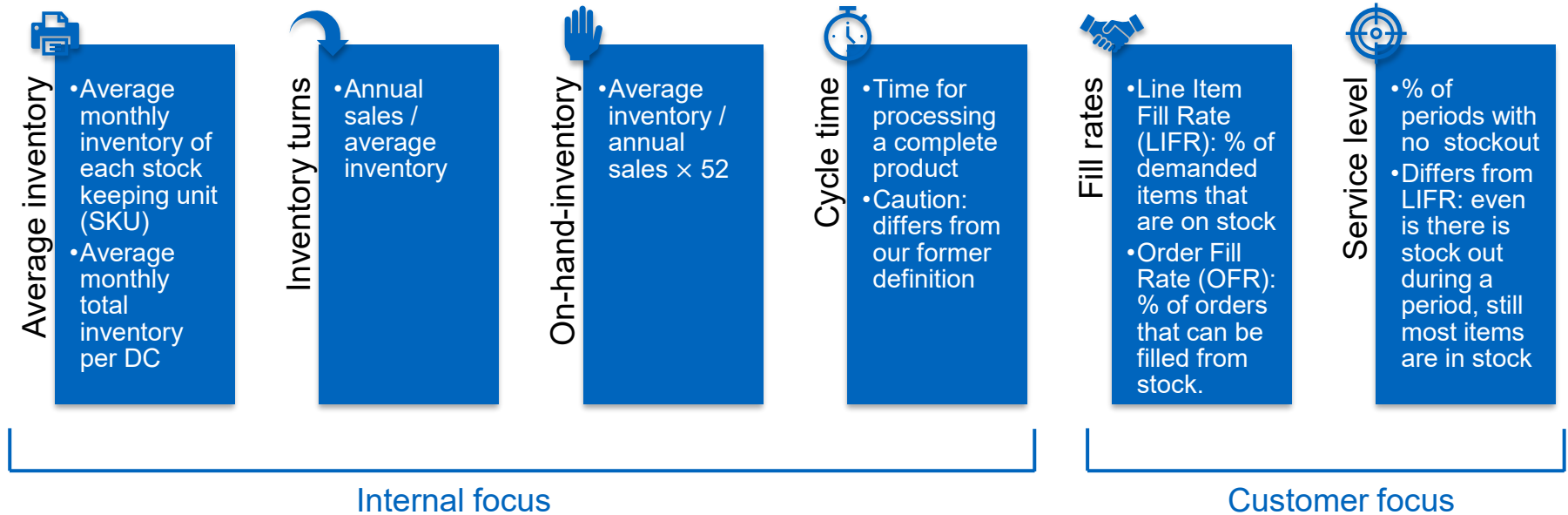
Internal capabilities

- More experience (learning curve effects)
- No more ramp-up problems
- Less demand uncertainty
- Superior processes are becoming key success indicators

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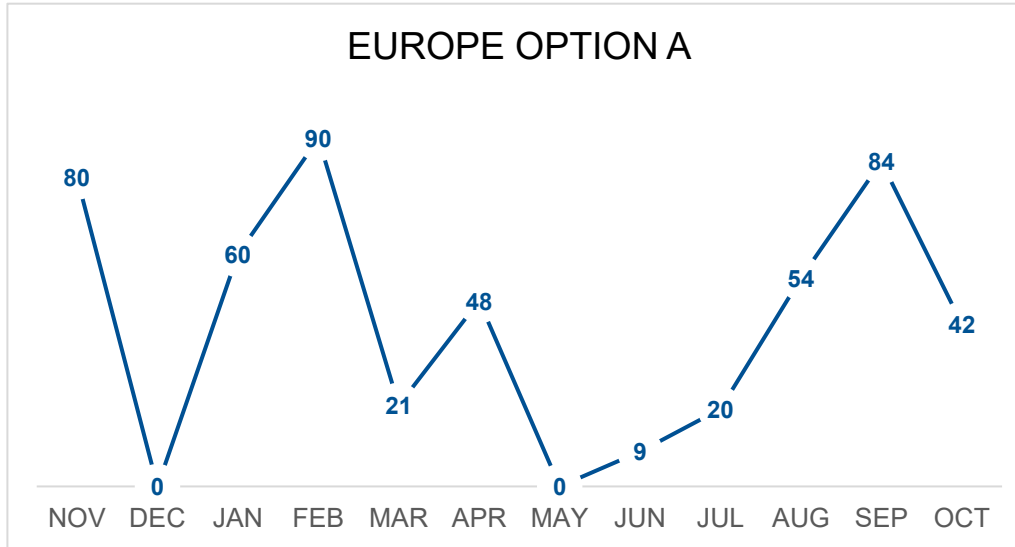
3. KPIs/Metrics for HP DeskJet Supply Chain



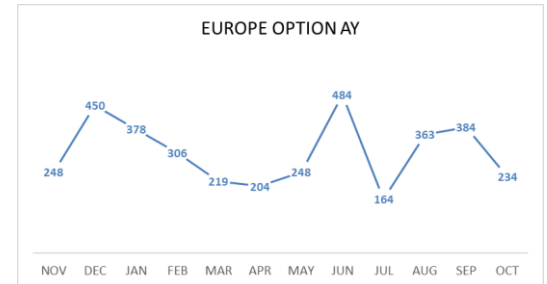
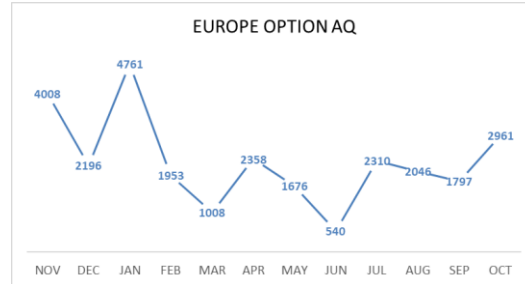
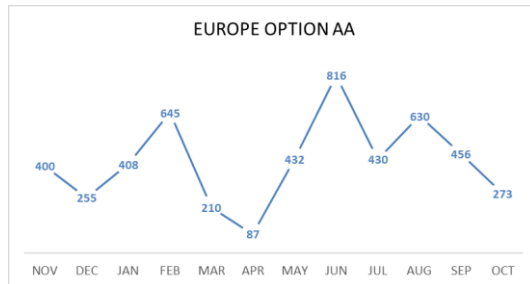
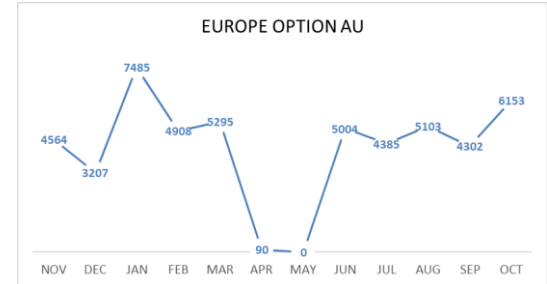
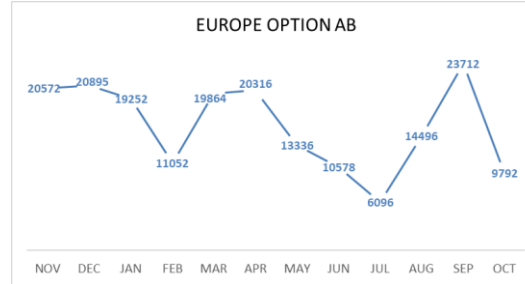
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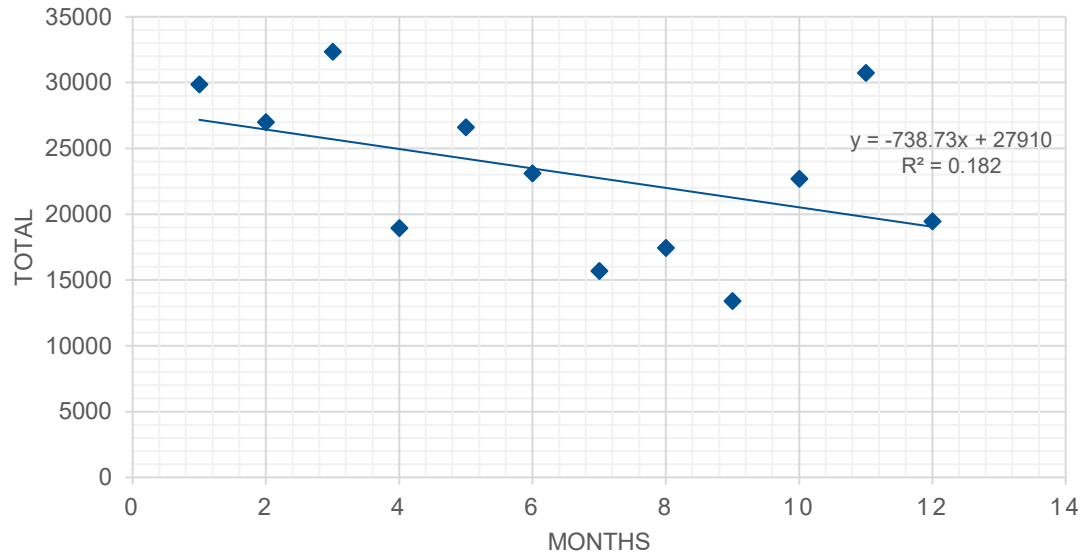
4. Demand at European DC



4. Demand at European DC



4. Demand at European DC



$F(df_1 = 1, df_2 = 10) = 2.224$
 $p = 0.166 > 0.1$
Hence, no trend

4. Demand at European DC

Recall: Under a normal distribution, mean equals median, and there are about 68% of all values within a range of 2 standard deviations and about 95% of values within a range of 4 standard deviations

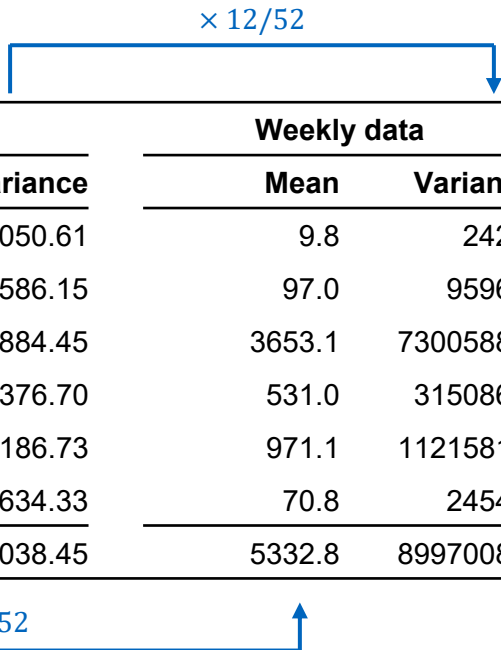
Europe Option	Mean	Median	Normalized gap*	Values within range of 2 standard deviations	values within range of 4 standard deviations
A	42.3	45	0.08	50%	100%
AA	420.2	419	0.01	58%	100%
AB	15830.1	16874	0.19	75%	100%
AQ	2301.2	2121	0.15	67%	92%
AU	4208.0	4736	0.24	75%	100%
AY	306.8	277	0.29	75%	100%
TOTAL	23108.6	22897.5	0.03	58%	100%

This suggests
normality of
demand

*) $|\text{mean} - \text{median}| / \text{standard deviation}$

4. Demand at European DC

Break down into weekly data for subsequent analysis.

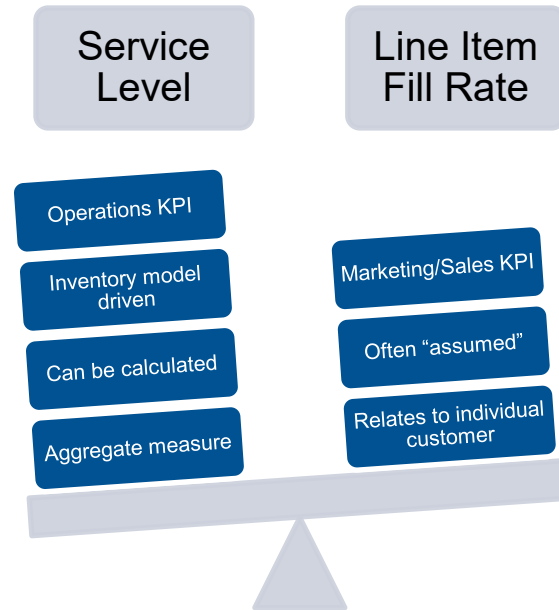


Europe Option	Monthly data			Weekly data	
	Mean	Std. dev	Variance	Mean	Variance
A	42.3	32.4	1050.61	9.8	242.4
AA	420.2	203.9	41586.15	97.0	9596.8
AB	15830.1	5624.6	31635884.45	3653.1	7300588.7
AQ	2301.2	1168.5	1365376.70	531.0	315086.9
AU	4208.0	2204.6	4860186.73	971.1	1121581.6
AY	306.8	103.1	10634.33	70.8	2454.1
TOTAL	23108.6	6244.0	38987038.45	5332.8	8997008.9

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5. Service Level (SL) and Line Item Fill Rate (LIFR)



5. Service Level (SL) and Line Item Fill Rate (LIFR)

Note: this is almost the same slide we had in Session 6, the order-up-to model

Let Demand be normally distributed with parameters μ and σ . Let $z = \frac{Q - \mu}{\sigma}$, where Q is the order quantity. Further, let ϕ denote the density of a standard normal distribution and let Φ denote the corresponding cumulative distribution function.

Observe that $\phi'(z) = \frac{d}{dz} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) = -z \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) = -z\phi(z)$. Expected lost sales can be expressed as $L(z) := \int_z^\infty (x - z)\phi(z)dx = -\int_z^\infty x\phi(z)dx - z \int_z^\infty \phi(z)dx = \phi(z) - z(1 - \Phi(z))$.

With that, expected sales, $E(z) := \mu - \sigma L(z)$.

Finally,
$$LIFR(z) := \frac{E(z)}{\mu} = 1 - \frac{\sigma L(z)}{\mu} = 1 - \sigma \frac{\phi(z) - z(1 - \Phi(z))}{\mu}.$$

It suffices to remember this formula.
Or to remember, where to find it.

5. Service Level (SL) and Line Item Fill Rate (LIFR)

Assumption 1: Annual holding costs per year are 48% of item value (between 12% and 60%, see page 8).

Assumption 2: Assumed per-unit margin is 22.4%, that is, four times HP's corporate margin (5.6%).

Assumption 3: Per unit backlogged, 50% of the per-unit margin is lost.

Let V be the value of a printer. Hence,

$$c_o = \frac{0.48}{52} V = 0.0092V$$

$$c_u = 0.224 \times 0.5V = 0.112V$$

One period = 1 week

$$SL = \frac{c_u}{c_u + c_o} = \frac{.112V}{0.0092V + 0.112V} = 92.4\%$$

5. Service Level (SL) and Line Item Fill Rate (LIFR)

Consider the service level of $SL = 92.4\%$.

Use the formula $LIFR(z) := \frac{E(z)}{\mu} = 1 - \sigma \frac{L(z)}{\mu} = 1 - \sigma \frac{\phi(z) - z(1 - \Phi(z))}{\mu} \approx 1 - 0.034 \frac{\sigma}{\mu}$.

Recall, in this formula μ and σ are the parameters of the normally distributed demand. Hence, $LIFR(z)$ depends on the actual SKU, in this case, the corresponding Europe Option.

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6. Safety-Stock (order-up-to model) Baseline Scenario

Lead time 5 weeks + cycle length 1 week = **6 weeks**

Each week, order this amount minus on-hand inventory
+ backorders – in-transit inventory

Product	Mean	Std.Dev.	Order-up-to level (S)	LIFR	Safety-Stock
A	58.6	38.1	113.2	97.77%	55.9
AA	581.8	240.0	925.2	98.59%	351.7
AB	21918.6	6618.4	31392.0	98.97%	9699.9
AQ	3186.2	1375.0	5154.3	98.52%	2015.1
AU	5826.5	2594.1	9539.6	98.48%	3801.9
AY	424.8	121.3	598.5	99.02%	177.8
TOTAL	31996.5		47723.0		

Expected inventory level
when new order arrives.

Similar to 98% “developed” by marketing.

6. Safety-Stock (order-up-to model) Baseline Scenario

Detailed split-up into different forms of inventory and expected backorders

Product	cycle inventory	in-transit inventory	Expected backorders	safety stock (left over inventory)	Total
A	9.8	48.8	1.3	55.9	
AA	97.0	484.8	8.2	351.7	
AB	3653.1	18265.5	226.4	9699.9	
AQ	531.0	2655.2	47.0	2015.1	
AU	971.1	4855.4	88.7	3801.9	
AY	70.8	354.0	4.2	177.8	
Weekly average	2666.4	26663.8		16102.3	45432.4
sum			375.8		375.8

Assuming linear consumption. To calculate this value, add up all entries above and divide by 2, as in the EOQ model; this is not the EOQ model, but the same line of reasoning applies

6. Safety-Stock (order-up-to model) Baseline Scenario

Total annual inventory holding costs per \$ unit price:	21,800
Total annual inventory backorder costs per \$ unit price:	2200
Total demand:	277,300

Assume unit price of \$400 (does not really matter though):

Total inventory related costs:	\$ 9,598,000
Total revenue:	\$ 110,921,000

← As per case: \$400,000,000 sales, 600,000 units, which would lead to \$667; hence, \$400 is a more conservative approach.

Share of inventory costs:	8.65 %
Assumed margin:	22.4 %

← Even only 8.65% of revenue matter!

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7. Safety-Stock (order-up-to model) Air Shipments

Lead time **1 week** + cycle length 1 week = **2 weeks**

Product	Mean	Std.Dev.	Order-up-to level (S)	LIFR	Safety-Stock
A	19.5	22.0	51.1	96.14%	32.3
AA	193.9	138.5	392.2	97.56%	203.0
AB	7306.2	3821.1	12775.7	98.21%	5673.3
AQ	1062.1	793.8	2198.4	97.44%	1203.9
AU	1942.2	1497.7	4086.0	97.36%	1776.1
AY	141.6	70.1	241.9	98.31%	171.1
TOTAL	10665.5		19745.2		

Note: these are somewhat smaller, since “averaging out” over weeks disappears.

7. Safety-Stock (order-up-to model) Air Shipments

Detailed split-up into different forms of inventory and expected backorders

Product	cycle inventory	in-transit inventory	Expected backorders	safety stock (left over inventory)	Total
A	9.8	9.8	0.8	32.3	
AA	97.0	97.0	4.7	203.0	
AB	3653.1	3653.1	130.7	5673.3	
AQ	531.0	531.0	27.2	1203.9	
AU	971.1	971.1	51.2	1776.1	
AY	70.8	70.8	2.4	171.1	
Weekly average	2666.4	5332.8		9059.7	19962.2
sum			217		217

← About half the inventory

7. Safety-Stock (order-up-to model) Air Shipments

Total annual inventory holding costs per \$ unit price:	8,300
Total annual inventory backorder costs per \$ unit price:	1,263
Total demand:	277,300

Assume unit price of \$400 (does not really matter though):

Total inventory related costs:	\$ 3,826,000
Total revenue:	\$ 110,921,000

Share of inventory costs:	3.45 %
Assumed margin:	22.4 %

Savings per printer:	\$20.80
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Would this offset extra
transport cost?

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8. Safety-Stock (order-up-to model) Pooling + Ship

Lead time 5 weeks + cycle length 1 week = **6 weeks**

Product	Mean	Std.Dev.	Order-up-to level (S)	LIFR	Safety-Stock
POOLED	31996.5	7347.2	42513.2	99.21%	251.3

8. Safety-Stock (order-up-to model) Pooling + Ship

Detailed split-up into different forms of inventory and expected backorders

Product	cycle inventory	in-transit inventory	Expected backorders	safety stock (left over inventory)
POOLED	2666.4	26663.8	251.3	10768.0

8. Safety-Stock (order-up-to model) Pooling + Ship

Total annual inventory holding costs per \$ unit price:	19,250
Total annual inventory backorder costs per \$ unit price:	1,460
Total demand:	277,300

Assume unit price of \$400 (does not really matter though):

Total inventory related costs:	\$ 8,284,000
Total revenue:	\$ 110,921,000

Share of inventory costs:	7.47 %
Assumed margin:	22.4 %

← Increases margin by 1.18 percentage points, profit increase of 5.3%.

Savings per printer:	\$4.74
Improvement of LIFR:	98.8% to 99.21%

For printers worth \$667, this corresponds to about \$7.90

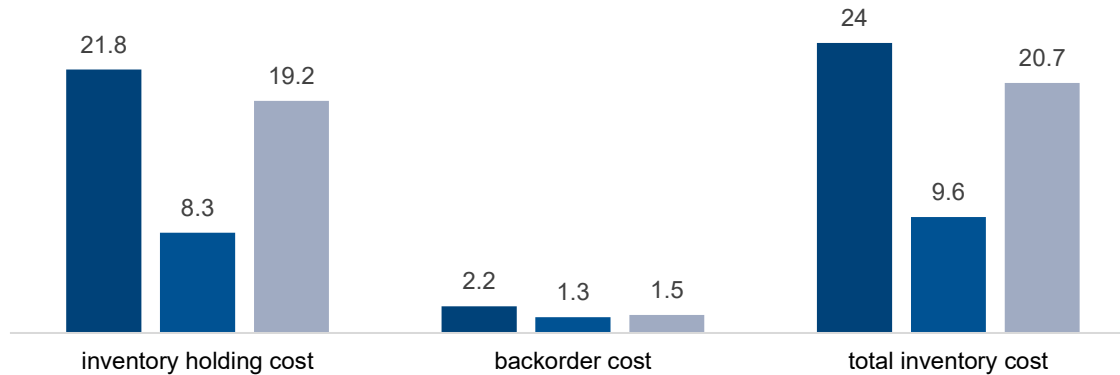
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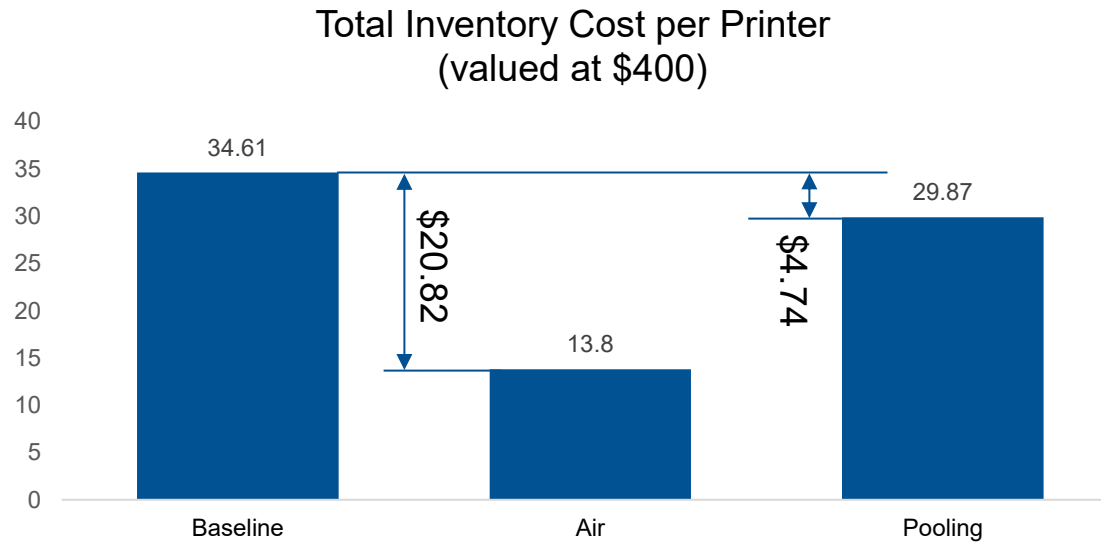
9. Alternatives and Implementation

Total annual inventory cost in thousands per \$ unit value

■ Baseline ■ Air ■ Pooling



9. Alternatives and Implementation



9. Alternatives and Implementation

Adding more complexity?

How to keep processes lean?

What is the value of lean?

Process redesign and postponement?

9. Recommended Solution

Postponement and product redesign

- Consider one standard printer per box
- Include manual with all languages
- Include all possible European plugs

Possible costs? → Perhaps about \$ 0.50, possible savings, about \$5.00, depending on further assumptions above. Certainly a strong way to reduce costs, increase LIFR and keep processes lean and simple.